

ENHANCING SEGMENTATION APPROACHES FROM GAUSSIAN MIXTURE MODEL AND EXPECTED MAXIMIZATION TO SUPER PIXEL DIVISION ALGORITHM

Christo Ananth¹, D.R.Denslin Brabin²

¹College of Engineering, AMA International University, Kingdom of Bahrain

²Computer Science and Engineering, Madanapalle Institute of Technology and Science, Andhra Pradesh, India

Abstract

Automatic liver tumor segmentation would bigly influence liver treatment organizing strategy and follow-up assessment, as a result of organization and joining of full picture information. Right now, develop a totally programmed technique for liver tumor division in CT picture. Introductory liver division comprises of applying a functioning form strategy. In the wake of separating liver applying Super pixel division Algorithm for portioning liver tumor proficiently. In the proposed work, we will investigate these procedures so as to improve division of various segments of the CT pictures. The exploratory outcomes indicated that the proposed strategy was exact for liver tumor division.

Keywords: Active pixel, Gaussian mixture model, Cellular Neural Networks, Level set, Expectation-maximization algorithm.

I. INTRODUCTION

As indicated by the World Health Organization, liver malignancy was the second most basic reason for disease instigated passings. Hepatocellular carcinoma (HCC) is the most widely recognized sort of essential liver malignancy which is the 6th most pervasive disease [20]. What's more, the liver is likewise a typical site for optional tumors. Liver treatment arranging strategies would benefit from an exact and quick injury division that takes into consideration resulting assurance of volume-and surface based data. Also, having an institutionalized and programmed division technique would encourage a progressively solid treatment reaction grouping. Liver tumors show a high changeability in their shape,



appearance and restriction. They can be either hypodense (seeming darker than the encompassing solid liver parenchyma) or hyperdense (seeming more brilliant), and can also have an edge because of the difference operator aggregation, calcification or corruption. The individual appearance relies upon injury type, state, imaging (hardware, settings, differentiate technique and timing), and can fluctuate significantly from patient to tolerant. This high changeability makes liver sore division a difficult undertaking practically speaking.

Liver malignant growth is one of the diseases which has the most elevated death rate on the planet. Every year, the dismalness and mortality of liver disease increment consistently. Processed tomography (CT) is the generally utilized imaging technique for screening, diagnosing, arranging, and even guess appraisal of liver malignancy. Division of liver injuries can assist doctors with diagnosing malignant growth and make proper treatment alternatives with increasingly advantageous, and can likewise rapidly survey the adequacy of careful treatment. In any case, physically estimating the tumor from an enormous number of CT cuts is very tedious and depends on the experience of the doctor, which is a lot of emotional and helpless to impedance from information contrasts among various doctors. Along these lines, it is exceptionally encouraging to plan a goal and exact plan for programmed liver tumor division to assist doctors with bettering decipher CT pictures. In any case, in light of the fact that the number, area, size and state of liver tumors are essentially extraordinary among the patients, also, the tumor limit is constantly obscured and the complexity among tumor and its encompassing tissues is low, precise division of liver tumor is as yet a troublesome assignment.

So as to address this issue, specialists have put resources into this investigation and proposed numerous strategies. During the previous barely any decades, they for the most part centered around creating calculations, for example, level set, watershed, factual shape model, district developing, dynamic form model, edge handling, chart cuts and customary AI techniques that require physically extricate tumor highlights. There are numerous famous superpixel approaches, for example, standardized cut [23], SLIC [24], LSC [25], ERS [26], SEEDS [27], mean move [28], Turbo-pixel [29], graphcuts [30], and



pseudoboolean improvement (PB) [31]. Every calculation has its own bit of leeway and inconvenience for superpixel division, notwithstanding, it is still testing to build up a high caliber and ongoing superpixel calculation that displays the properties including great limit adherence, smaller requirements, normal shapes and low computational intricacy. So as to fulfill these ideal prerequisites, we propose a constant superpixel division calculation by Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [32] to accomplish preferred execution over best in class techniques. This is the first occasion when that the DBSCAN grouping calculation is applied to superpixel division. An ideal superpixel strategy needs to not just satisfy the prerequisite of good limit adherence, yet in addition be proficient. Since the superpixels are utilized as a preprocessing step in vision applications, the calculation of great superpixels with less calculation is liked. Right now, propose another continuous superpixel technique utilizing DBSCAN [32] bunching, which acquires the upsides of existing superpixel calculations, (for example, SLIC) and further conveys forward past these favorable circumstances. Our DBSCAN superpixel division calculation isn't just more productive yet additionally more precise than past superpixel calculations.

II. RELATED WORK

For instance, Zhou et al. [1] proposed a bound together level set strategy (LSM) for liver tumor division. They utilized neighborhood pixel force bunching joined with shrouded Markov arbitrary field to develop a bound together LSM. At that point, local data and edge data were utilized to secure the tumor form, with the goal that the issue of edge spillage can be explained. Yan et al. [2] proposed a self-loader division strategy dependent on watershed change. They previously put seed focuses in the tumor territory physically as markers, at that point watershed change was performed to delineate and remove the tumor form in the picture. Along these lines, the thickness data of tumor can be procured as edge to isolate liver injury from its local tissues. At that point, refine the edge from the division sores for precise outcomes. Massoptier et al. [3] proposed a programmed liver tumor division calculation dependent on factual shape model, which utilized the dynamic form method of



inclination vector stream to get smooth and regular liver tumor division results without the need of cooperation. Wong et al. [4] proposed a self-loader tumor division strategy dependent on area developing technique. They first sketch the locale of enthusiasm of tumor physically and ascertain the seed point and highlight vector in it, at that point territorial developing calculation was performed to check the tumor voxels. Fusing information based imperatives into the developing technique guarantees the portioned tumor size and shape are inside a sensible range. Yim et al. [5] proposed a division strategy dependent on dynamic form model, which acknowledged liver sore division by substituting manual introduction and circular instatement of the dynamic shape. Through this technique, they at last got a decent division result. Park et al. [6] proposed a factual ideal thresholding strategy for tumor division. They use strategies as most extreme back choice and paired morphological sifting to section the liver, at that point utilizing calculation, for example, blended likelihood thickness and least absolute likelihood mistake to compute the ideal edge. At long last, the division of liver tumor is accomplished by playing out the determined edge. Linguraru et al. [7] proposed a technique dependent on chart cut improvement. They utilized the shape parameterization technique to recognize the liver movement forms, which can address the accompanying anomalous tumor division circumstance. The tumor division is then acknowledged by shape obliged based diagram cut calculation. As of late, conventional AI based picture division techniques have assumed a functioning job in liver tumor division situations. The vast majority of these techniques need to physically plan tumor include extraction strategies, and building up a model to prepared the highlights, making the model can recognize tumor pixels. Huang et al. [8] proposed a liver tumor division strategy dependent on extraordinary learning machine (ELM), which groups tumor and non-tumor voxels via preparing classifiers in stochastic element subspace sets. The ELM is chosen as the essential classifier for model preparing. To additionally improve the division execution, casting a ballot instrument is acquainted with choose the last order aftereffects of these essential classifiers. Zhou et al. [9] proposed bolster vector machine (SVM) based technique for tumor division. They initially prepared a SVM model to section tumor district from a solitary cut and extricated its form through morphological activity. At that point, the shape is anticipated to the neighboring cuts for



next resampling, learning, and grouping. Rehashing this procedure until all cuts are prepared. Most AI based techniques can accomplish preferable execution over conventional ones, yet they are as yet hard to become familiar with the exact tumor highlight and powerless to information changes.

As of late, profound learning has entered into an assortment of uses and outperformed the cutting edge execution in numerous fields, for example, picture identification, arrangement, and division [10–15], which likewise energizes us to utilize this procedure in the liver tumor division task. Numerous specialists have just utilized profound learning strategies to investigate the errand of liver tumor division. Li et al. [16] proposed a H-DenseUNet which comprises of 2D UNet and 3D UNet for liver tumor division. The 2D UNet is utilized to separate tumor includes in a solitary cut while the 3D UNet is intended to gain proficiency with the spatial data of tumor between cuts. By structuring a blended element combination layer to mutually improve the element portrayal among intra-and between cut, they at long last procured an acceptable outcome in the liver tumor division challenge. Christ et al. [17] proposed a strategy dependent on falling two completely convolutional neural systems (FCN). They originally prepared a FCN to portion the liver from the stomach pictures and utilized the fragmented liver locale as contribution of the subsequent division organize, at that point the tumor division results can be gained constantly FCN. At last, they utilized 3D contingent arbitrary field to improve the tumor division results. Sun [18] et al. proposed a multi-channel completely convolutional organize (MC-FCN) based liver tumor division technique. They planned a MC-FCN to prepare differentiate improved CT pictures at various imaging stage because of each period of the information gives one of a kind data about the obsessive highlights of tumor. In particular, they performed organize preparing for each stage information in a solitary channel, and afterward blended the significant level component layer in various channels to acknowledge multi-channel preparing procedure. Exploratory outcomes showed the planned system can catch increasingly far reaching tumor includes and improve the model execution in some degree. Yuan et al. [19] built up a progressive structure of profound convolution-deconvolution neural system (CDNN) for division of liver tumors. They



initially prepared the CDNN model to fragment the liver locale from whole stomach CT succession, at that point they performed histogram leveling upgrade on the liver area, and view it as contribution of the following CDNN for tumor division. They supplanted the misfortune work by utilizing Jaccard separation during the preparation procedure to take out the example reweighting impact and accomplished an exceptional division results.

The staying of this article is sorted out as follows. In Section 3, we portray the subtleties of the proposed structure. The outcomes are introduced in Section 4, and finishing up in Section 5.

III. PROPOSED METHOD

a. PRE-PROCESSING

CT images usually contain many noises so we need to do smoothing for the image which makes the intensity distribution of liver to be smoothed too, so pre-processing step is necessary for accurate liver segmentation. The morphological operation is takes place to extract the mask image from binary Image.

b. LIVER SEGMENTATION

ACTIVE CONTOUR

The Active Contour system is especially designed for detecting blood vessels on coronary angiogram images for medical applications. The environment where these blood vessel structures are found contains materials such as segmented tissues and fluid debris. These become the unwanted background information that hinders the performance of the active contour in tracking the object outline accurately. In addition, angiogram images also contribute various obstructions that reduce the performance of the active contour such as color, noise, low contrast and poor quality images. Therefore, the Active Contour System developed in this chapter aims at optimizing the performance of object tracking by introducing various image processing techniques. An parametric active contour is simply a



set of contour points $[X(s), Y(s)]$ parameterized by $s \in [0, 1]$. Typically, parametric active contours are implemented by finding the contour that minimizes (1)

$$E = E_{\text{internal}} + E_{\text{external}}, \quad (1)$$

where E_{internal} is the internal energy of the active contour depicted in (2)

$$E_{\text{internal}} = \frac{1}{2} \int_0^1 \left\{ \alpha \left(\frac{dX(s)}{ds} \right)^2 + \left(\frac{dY(s)}{ds} \right)^2 + \beta \left[\left(\frac{d^2X(s)}{ds^2} \right)^2 + \left(\frac{d^2Y(s)}{ds^2} \right)^2 \right] \right\} ds \quad (2)$$

Here, α and β are two nonnegative weighting parameters expressing respectively the degree of the resistance to stretching and bending of the contour. The external energy E_{external} is typically defined such that the contour seeks the edges in the image, (3),

$$E_{\text{external}} = - \int_0^1 f(X(s), Y(s)) ds \text{ where } f(x, y) = |\nabla I(x, y)|^2 \quad (3)$$

Active contours can be characterized as the procedure to acquire deformable models or structures with limitations and powers in a picture for division. Shape models portray the article limits or some other highlights of the picture to frame a parametric bend or form. Ebb and flow of the models is resolved with different shape calculations utilizing outer and inside powers applied. Vitality practical is constantly connected with the bend characterized in the picture. Outer vitality is characterized as the blend of powers because of the picture which is explicitly used to control the situating of the form onto the picture and inner vitality, to control the deformable changes [3]. Limitations for a specific picture in the form division rely upon the necessities. The ideal form is acquired by characterizing the base of the vitality useful. Misshaping of the form is portrayed by an assortment of focuses that finds a shape. This form fits the necessary picture shape characterized by limiting the vitality practical.

c. TUMOR SEGMENTATION

SUPERPIXEL SEGMENTATION

Superpixel is a group of pixels in proximity that has similar intensity. Simple Linear Iterative Clustering (SLIC) algorithm [20] is applied due to its fast computational time [21],



[22]. The size of original superpixels extracted from SLIC is different as there might be a small number of pixels near each other with the similar pixel value in some of the region of the image (most of the time in tumor region), while in the non-tumor part of the image their size will be large. However, we need the same size of superpixels in order to apply PCA. This problem is solved as follows:

We computed average size of the superpixel as shown in equation (4)

$$M = \frac{1}{N} \sum_{i=1}^N n_i \quad (4)$$

Where N is the number of superpixels, n_i is number of pixels in i th superpixel, M is average number of pixels. Then, the size of each superpixel is made same as of the average one by padding some pixel value to the smaller size superpixel and removing some intensity value from the large size superpixels. Instead of appending random intensity values to smaller size superpixels, we pad by repeating the last pixels value of the superpixel itself. Finally, the superpixel matrix is generated. Compute average superpixel. Determine the covariance of superpixels. Calculate the eigenvectors and eigenvalues of the covariance matrix. The superpixels onto Principal components that contain most variance of the data. Here, the number of Principal components is same as the number of superpixels. As stated, the eigenvectors or Principal components that contain at least 95% variance of superpixels can represent the whole image by confidence. This reduces the dimensional space as most of the information is contained in the first two or three largest eigenvalues. In our implementation, 95% variance of superpixels was contained in the top two principal components for most of the images. Calculate the distance of each superpixel in respect to average superpixel. While computing distance, we should consider the distribution of superpixels in the principal component coordinate system. To incorporate this concept we computed the distance along the principal components.

DBSCAN CLUSTERING

Clustering analysis or simply Clustering is basically an Unsupervised learning method that divides the data points into a number of specific batches or groups, such that



the data points in the same groups have similar properties and data points in different groups have different properties in some sense. It comprises of many different methods based on different evolution. Fundamentally, all clustering methods use the same approach i.e. first we calculate similarities and then we use it to cluster the data points into groups or batches. Here we will focus on Density-based spatial clustering of applications with noise (DBSCAN) clustering method.

Clusters are dense regions in the data space, separated by regions of the lower density of points. The DBSCAN algorithm is based on this intuitive notion of “clusters” and “noise”. The key idea is that for each point of a cluster, the neighborhood of a given radius has to contain at least a minimum number of points. Partitioning methods (K-means, PAM clustering) and hierarchical clustering work for finding spherical-shaped clusters or convex clusters. In other words, they are suitable only for compact and well-separated clusters. Moreover, they are also severely affected by the presence of noise and outliers in the data.

DBSCAN algorithm requires two parameters depicted in fig(1) , eps : It defines the neighborhood around a data point i.e. if the distance between two points is lower or equal to ‘eps’ then they are considered as neighbors. If the eps value is chosen too small then large part of the data will be considered as outliers. If it is chosen very large then the clusters will merge and majority of the data points will be in the same clusters. One way to find the eps value is based on the k-distance graph. MinPts: Minimum number of neighbors (data points) within eps radius. Larger the dataset, the larger value of MinPts must be chosen. As a general rule, the minimum MinPts can be derived from the number of dimensions D in the dataset as, $\text{MinPts} \geq D+1$. The minimum value of MinPts must be chosen at least 3.



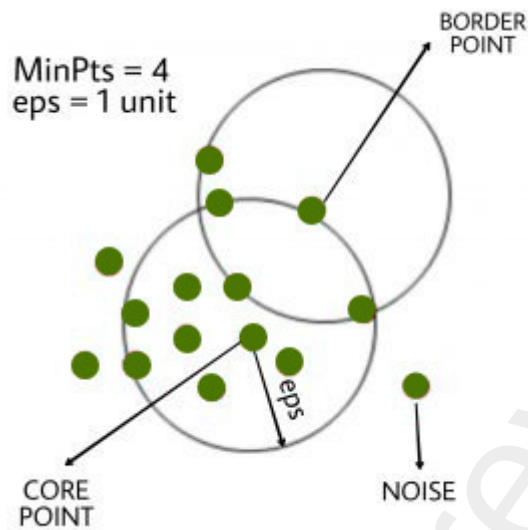


Fig 1: DBSCAN parameters

1. Find all the neighbor points within ϵ and identify the core points or visited with more than MinPts neighbors.
2. For each core point if it is not already assigned to a cluster, create a new cluster.
3. Find recursively all its density connected points and assign them to the same cluster as the corepoint.
A point a and b are said to be density connected if there exist a point c which has a sufficient number of points in its neighbors and both the points a and b are within the ϵ distance. This is a chaining process. So, if b is neighbor of c , c is neighbor of d , d is neighbor of e , which in turn is neighbor of a implies that b is neighbor of a .
4. Iterate through the remaining unvisited points in the dataset. Those points that do not belong to any cluster are noise.



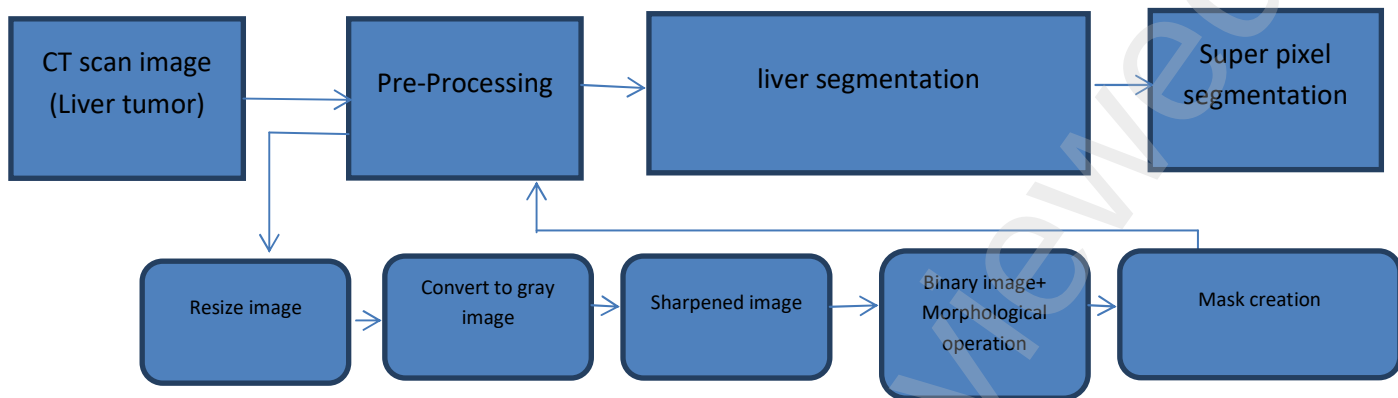


Fig.2. Block Diagram Of Proposed System

The CT scan image is pre-processed through a morphological operation where the image gets sharpened and converted into a binary image. At the end of preprocessing, a mask image is extracted. Then active contour segmentation is applied to extract the liver region from other parts of the human body. Through superpixel segmentation, the liver tumor is determined and then DBSCAN clustering is used to cluster the output image. Thus the process is used to segment the liver tumor from the healthy tissues of the liver which is illustrated in fig 2.



IV. RESULTS AND DISCUSSION

The result of the liver segmentation through super pixel segmentation is evaluated and corresponding output is depicted in fig 3. The Input image is undergoes in preprocessing, liver segmentation and tumor segmentation. Finally a ground truth image is obtained.

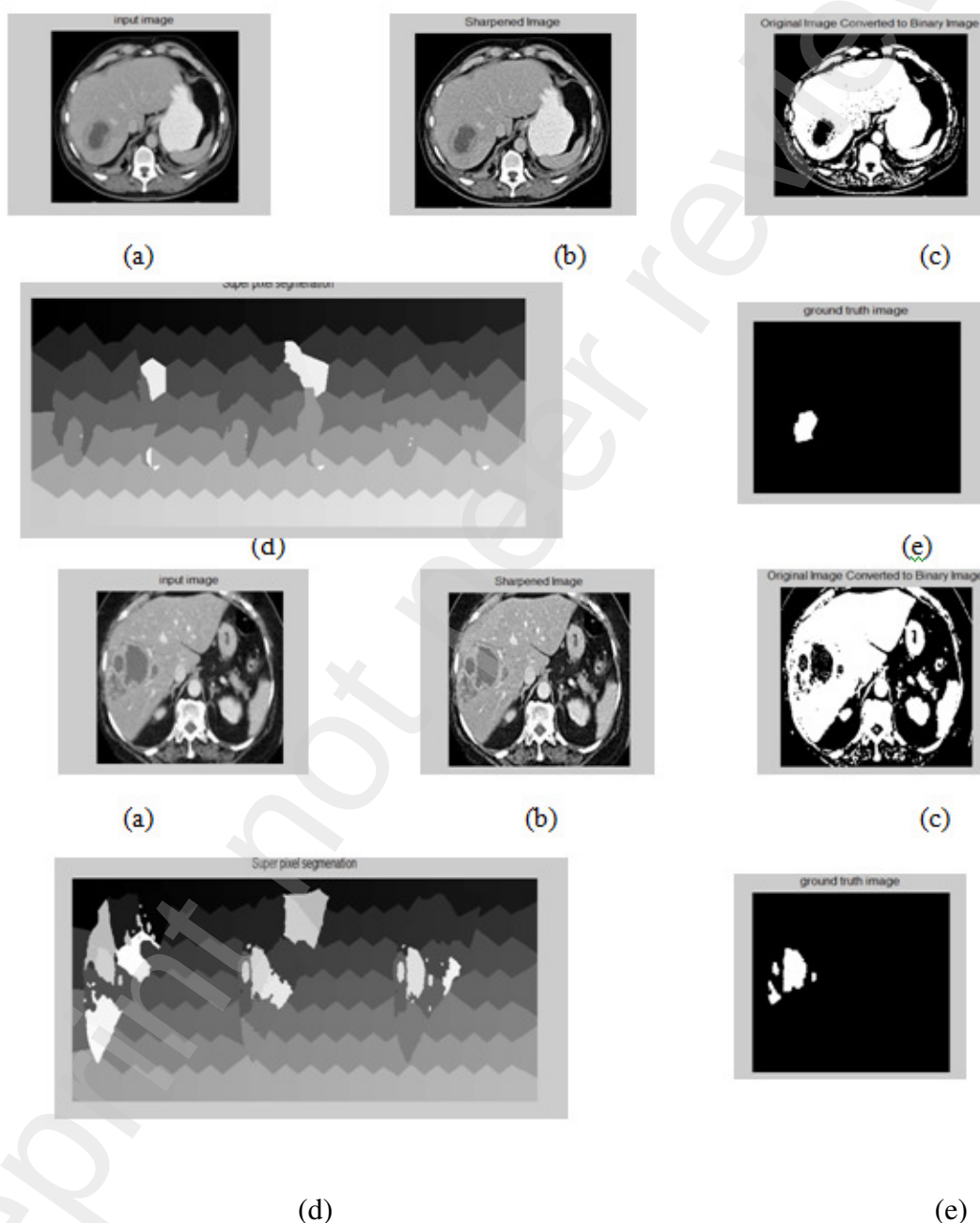


Fig 3: Segmentation of Liver tumor using Super pixel segmentation



(a) Input Image (b)Sharpened Image (c)Binary Image(d)Super Pixel Segmentation (e)ground truth image

TABLE I
PREDICTION OF LIVER TUMOR

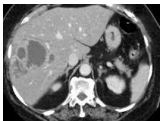
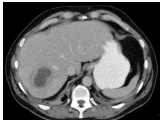
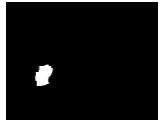
Input	Ground Truth	TP	TN	FP	FN
		1224	63521	716	75
		924	63968	34	610
		4614	60117	234	571



TABLE II

EVALUATION OF ACCURACY, SPECIFICITY, PRECISION, F-MEASURE, MCC, DICE JACCARD FOR THE LIVER SEGMENTATION.

S.NO	Accuracy	Sensitivity	Specificity	Precision	F-measure	MCC	Dice	Jaccard
1	0.9879	0.9423	0.9889	0.6309	0.7558	0.7657	0.7558	0.6074
2	0.9902	0.6023	0.9995	0.9645	0.7416	0.7581	0.7416	0.5893
3	0.9877	0.8899	0.9961	0.9517	0.9198	0.9137	0.9198	0.8514

The liver segmentation output is predicted and calculated the true positive, true negative, false positive and false negative value for each input image is illustrated in table 1. The accuracy, specificity, precision, F-measure, MCC, Dice Jaccard for the liver segmentation is carried on and compared with the existing system. Our proposed method provides a high accuracy than existing methods is represented in table 2.

V. CONCLUSION

In this paper, we developed Superpixel based liver tumors segmentation from abdominal CT volumes. Novel picture Active contour division, superpixel division calculation utilizing DBSCAN bunching for tumor division. The liver part is sectioned from the other piece of body through form division. Our proposed superpixel division first delivers the underlying superpixel results with the comparative hues by playing out the DBSCAN grouping calculation, and afterward consolidates the little beginning superpixels with their closest neighbor superpixels by thinking about their shading and spatial data. Our calculation accomplishes the best in class execution at a considerably littler calculation cost, and altogether beats the calculations that require increasingly computational expenses in any event, for the pictures including complex items or complex surface locales. In future work, we will acquire better minimization of



superpixels by building up another DBSCAN calculation that has the worldwide ideal property.

VI. SCOPE FOR FUTURE ENHANCEMENT

In a future work, we will research Hidden Markov Random Fields with Expected Maximization for bigger scale organize estimate, and consider the effective answer for the expanded effectiveness and lessened emphasis in a system with high portability. It is additionally a promising future work to match the accuracy and object delineation time of the existing system for multi-shape structures.

REFERENCES

- [1].Zheng Z, Zhang X, Xu H, Liang W, Zheng S, Shi Y. A unified level set framework combining hybrid algorithms for liver and liver tumor segmentation in ct images. BioMed Res Int. 2018; 2018. <https://doi.org/10.1155/2018/3815346>.
- [2].Yan J, Schwartz LH, Zhao B. Semiautomatic segmentation of liver metastases on volumetric ct images. Med Phys. 2015; 42(11):6283–93.
- [3].Massoptier L, Casciaro S. A new fully automatic and robust algorithm for fast segmentation of liver tissue and tumors from ct scans. Eur Radiol. 2008; 18(8):1658.
- [4].Wong D, Liu J, Fengshou Y, Tian Q, Xiong W, Zhou J, Qi Y, Han T, Venkatesh S, Wang S-c. A semi-automated method for liver tumor segmentation based on 2d region growing with knowledge-based constraints. In: MICCAI Workshop, vol. 41. Berlin: Springer-Verlag Berlin Heidelberg; 2008. p. 159.
- [5].Yim PJ, Foran DJ. Volumetry of hepatic metastases in computed tomography using the watershed and active contour algorithms. In: 16th IEEE Symposium Computer-Based Medical Systems, 2003. Proceedings. IEEE: 2003. p. 329–35. <https://doi.org/10.1109/cbms.2003.1212810>.
- [6].Park S-J, Seo K-S, Park J-A. Automatic hepatic tumor segmentation using statistical optimal threshold. In: International Conference on Computational Science. Springer: 2005. p. 934–40. https://doi.org/10.1007/11428831_116.



- [7].Linguraru MG, Richbourg WJ, Watt JM, Pamulapati V, Summers RM. Liver and tumor segmentation and analysis from ct of diseased patients via a generic affine invariant shape parameterization and graph cuts. In: International MICCAI Workshop on Computational and Clinical Challenges in Abdominal Imaging. Springer: 2011. p. 198–206. https://doi.org/10.1007/978-3-642-28557-8_25.
- [8].Huang W, Yang Y, Lin Z, Huang G. -B., Zhou J, Duan Y, Xiong W. Random feature subspace ensemble based extreme learning machine for liver tumor detection and segmentation. In: 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE: 2014. p. 4675–8. <https://doi.org/10.1109/embc.2014.6944667>.
- [9].Zhou J, Xiong W, Tian Q, Qi Y, Liu J, Leow WK, Han T, Venkatesh SK, Wang S-c. Semi-automatic segmentation of 3d liver tumors from ct scans using voxel classification and propagational learning. In: MICCAI Workshop, vol. 41. Berlin: Springer-Verlag Berlin Heidelberg: 2008. p. 43.
- [10]. Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. In: Advances in Neural Information Processing Systems: 2012. p. 1097–105. <https://doi.org/10.1145/3065386>.
- [11]. Xu Y, Du J, Dai L-R, Lee C-H. An experimental study on speech enhancement based on deep neural networks. IEEE Signal Process Lett. 2014; 21(1):65–68.
- [12]. Yu L, Chen H, Dou Q, Qin J, Heng P-A. Automated melanoma recognition in dermoscopy images via very deep residual networks. IEEE Trans Med Imaging. 2017; 36(4):994–1004.
- [13]. Shin H. -C., Roth HR, Gao M, Lu L, Xu Z, Nogues I, Yao J, Mollura D, Summers RM. Deep convolutional neural networks for computer-aided detection: Cnn architectures, dataset characteristics and transfer learning. IEEE Trans Med Imaging. 2016; 35(5):1285–98.
- [14]. Havaei M, Davy A, Warde-Farley D, Biard A, Courville A, Bengio Y, Pal C, Jodoin P-M, Larochelle H. Brain tumor segmentation with deep neural networks. Med Image Anal. 2017; 35:18–31.



- [15]. Li X, Chen H, Qi X, Dou Q, Fu C-W, Heng P-A. H-denseunet: Hybrid densely connected unet for liver and tumor segmentation from ct volumes. *IEEE Trans Med Imaging*. 2018; 37(12):2663–74.
- [16]. Christ PF, Ettlinger F, Grün F, Elshaera MEA, Lipkova J, Schlecht S, Ahmaddy F, Tatavarty S, Bickel M, Bilic P, et al. Automatic liver and tumor segmentation of ct and mri volumes using cascaded fully convolutional neural networks. 2017. arXiv preprint arXiv:1702.05970.
- [17]. Sun C, Guo S, Zhang H, Li J, Chen M, Ma S, Jin L, Liu X, Li X, Qian X. Automatic segmentation of liver tumors from multiphase contrast-enhanced ct images based on fcns. *Artif Intell Med*. 2017; 83:58–66.
- [18]. Yuan Y. Hierarchical convolutional-deconvolutional neural networks for automatic liver and tumor segmentation. 2017. arXiv preprint arXiv:1710.04540.
- [19]. Forner, A., Llovet, J. M. & Bruix, J. Hepatocellular carcinoma. *The Lancet* **379**, 1245–1255 (2012).
- [20]. Z. Wu, Z. Hu, and Q. Fan. Superpixelbased unsupervised change detection using multi-dimensional change vector analysis and svm-based classification. *ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci*. 2012;7:257-262.
- [21]. Z. Tian, L. Liu, Z. Zhang, J. Xue, and B. Fei. A supervoxel-based segmentation method for prostate mr images. *Medical Physics*. 2016.
- [22]. J. Shi and J. Malik, “Normalized cuts and image segmentation,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 8, pp. 888–905, Aug. 2000.
- [23]. R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, “SLIC superpixels compared to state-of-the-art superpixel methods,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.
- [24]. Z. Li and J. Chen, “Superpixel segmentation using linear spectral clustering,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)* Jun. 2015, pp. 1356–1363
- [25]. M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, “Entropy rate superpixel segmentation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 2097–2104



- [26]. D. Comaniciu and P. Meer, "Mean shift: A robust approach toward feature space analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 5, pp. 603–619, May 2002.
- [27]. A. Levinstein, A. Stere, K. N. Kutulakos, D. J. Fleet, S. J. Dickinson, and K. Siddiqi, "TurboPixels: Fast superpixels using geometric flows," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 12, pp. 2290–2297, Dec. 2009.
- [28]. P. F. Felzenszwalb and D. P. Huttenlocher, "Efficient graph-based image segmentation," *Int. J. Comput. Vis.*, vol. 59, no. 2, pp. 167–181, Sep. 2004.
- [29]. M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discovery Data Mining (KDD)*, 1996, pp. 226–231.
- [30]. Y. Zhang, R. Hartley, J. Mashford, and S. Burn, "Superpixels via pseudo-Boolean optimization," in *Proc. IEEE Int. Conf. Comput. Vis.*, Nov. 2011, pp. 1387–1394.

